Ritz Method — Gelfand - Fomin Problem 8.4

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1 Problem Statement

4. Use the Ritz method to find an approximate solution of the problem of minimizing the functional

$$J[y] = \int_0^2 (y'^2 + y^2 + 2xy)\,dx, \qquad y(0) = y(2) = 0,$$

and compare the answer with the exact solution.

2 Solution

2.1 Exact Solution

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Problem: Find the extremal of $J[y] = \int_0^2 y'^2 + y^2 + 2xy \, dx \quad y(0) = y(2) = 0$

Soln:

$$F(x,y,y') = y'^2 + y^2 + 2xy$$

$$\left.\begin{aligned} F_y &= 2y + 2x \\ F_{y'} &= 2y' \\ \tfrac{d}{dx} F_{y'} &= 2y'' \end{aligned}\right\} \Rightarrow 2y + 2x - 2y'' = 0 \Rightarrow y'' - y = x$$

Solve Euler's eqn

complementary soln

$$y'' - y = 0 \Rightarrow m^2 - 1 = 0 \Rightarrow m = \pm 1 \Rightarrow y_c = C_1 e^x + C_2 e^{-x}$$

particular soln

$$y_p = Ax + B \quad y_p' = A \quad y_p'' = 0$$

substitute and solve A, B

$$-Ax - B = x \Rightarrow A = -1,\ B = 0 \Rightarrow y_p = -x$$

general soln

$$y_g = C_1 e^x + C_2 e^{-x} - x$$

Solve for C_1, C_2

$$\left.\begin{aligned} y(0) = 0: &\quad 0 = C_1 + C_2 \Rightarrow C_2 = -C_1 \\ y(2) = 0: &\quad 0 = C_1 e^2 + C_2 e^{-2} - 2 \end{aligned}\right\} \Rightarrow 2 = C_1 e^2 - C_1 e^{-2} = C_1 (e^2 - e^{-2})$$

$$C_1 = \frac{2}{e^2 - e^{-2}} \qquad C_2 = -\frac{2}{e^2 - e^{-2}}$$

$$\boxed{y = \frac{2(e^x - e^{-x})}{e^2 - e^{-2}} - x = \frac{2 \sinh x}{\sinh 2} - x}$$

* We could have actually just used complementary soln of $y_c = C_1 \sinh(x) + C_2 \cosh(x)$, that would have probably been easier

We will define the julia function of this for comparison against the approximations

```
y_exact(x) = 2 * sinh(x) / sinh(2) - x
have_exact = !isnothing(y_exact)
```

2.2 Ritz Approximations

We will use the `RitzMethod` package

```
using RitzMethod
import Optim
import Plots

# Define problem and basis
integrand(x, y, y_prime) = y_prime^2 + y^2 + 2 * x * y
domain = Interval(0.0, 2.0)
full_basis = [x -> x^k * (2 - x) for k in 1:3]

function solve_ritz(n_terms)
    minimizefunctional(
        integrand, domain, full_basis[1:n_terms];
        num_quad_nodes=5,
        method=Optim.BFGS(),
        options=Optim.Options(iterations=50, g_abstol=1e-20, x_reltol=1e-8),
    )
end

results = [solve_ritz(n) for n in 1:3]
```

2.2.1 One-Term Approximation

```
println(summarize(results[1]))
```

```
RitzResult:
  Domain:          [0.0, 2.0]
  Coefficients:    [-0.35714285714285726]
  Minimized value: -0.47619047619047633
  Converged:       true
```

2.2.2 Two-Term Approximation

```
println(summarize(results[2]))
```

```
RitzResult:
  Domain:          [0.0, 2.0]
  Coefficients:    [-0.20496894409937896, -0.15217391304347833]
  Minimized value: -0.5167701863354041
  Converged:       true
```

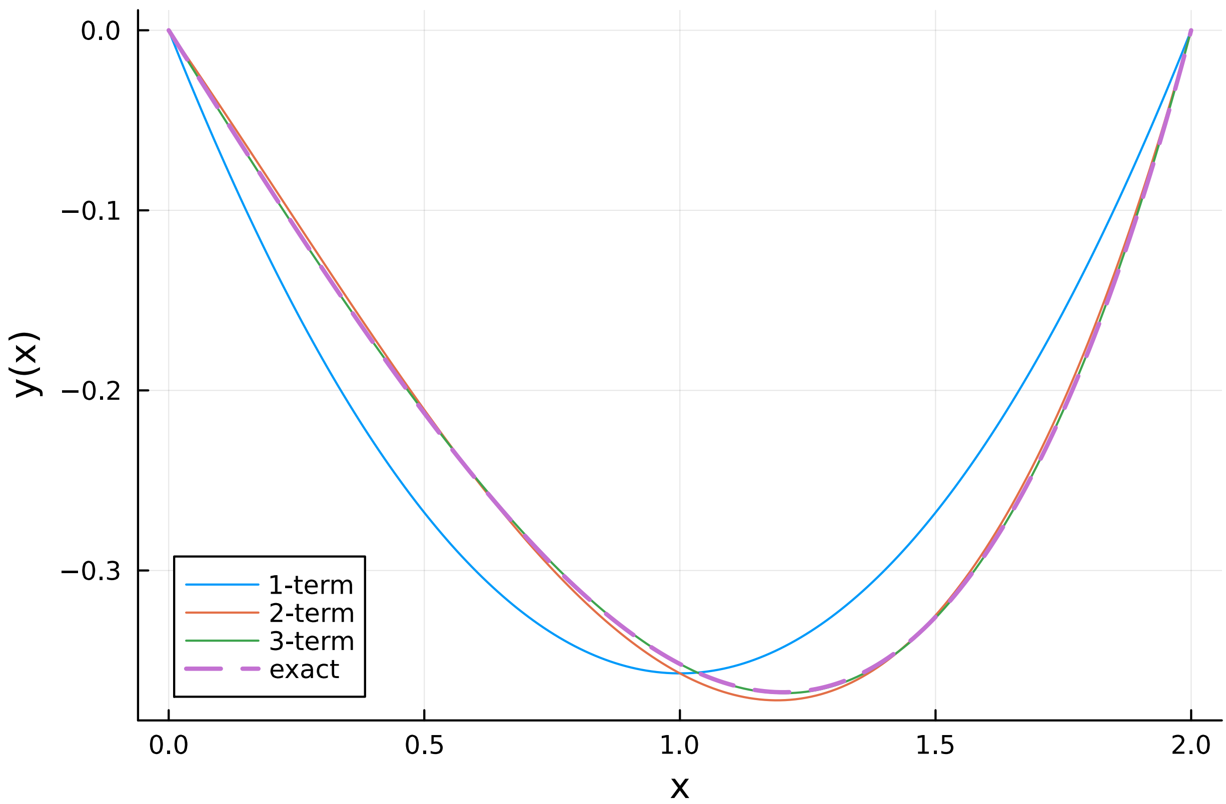
2.2.3 Three-Term Approximation

```
println(summarize(results[3]))
```

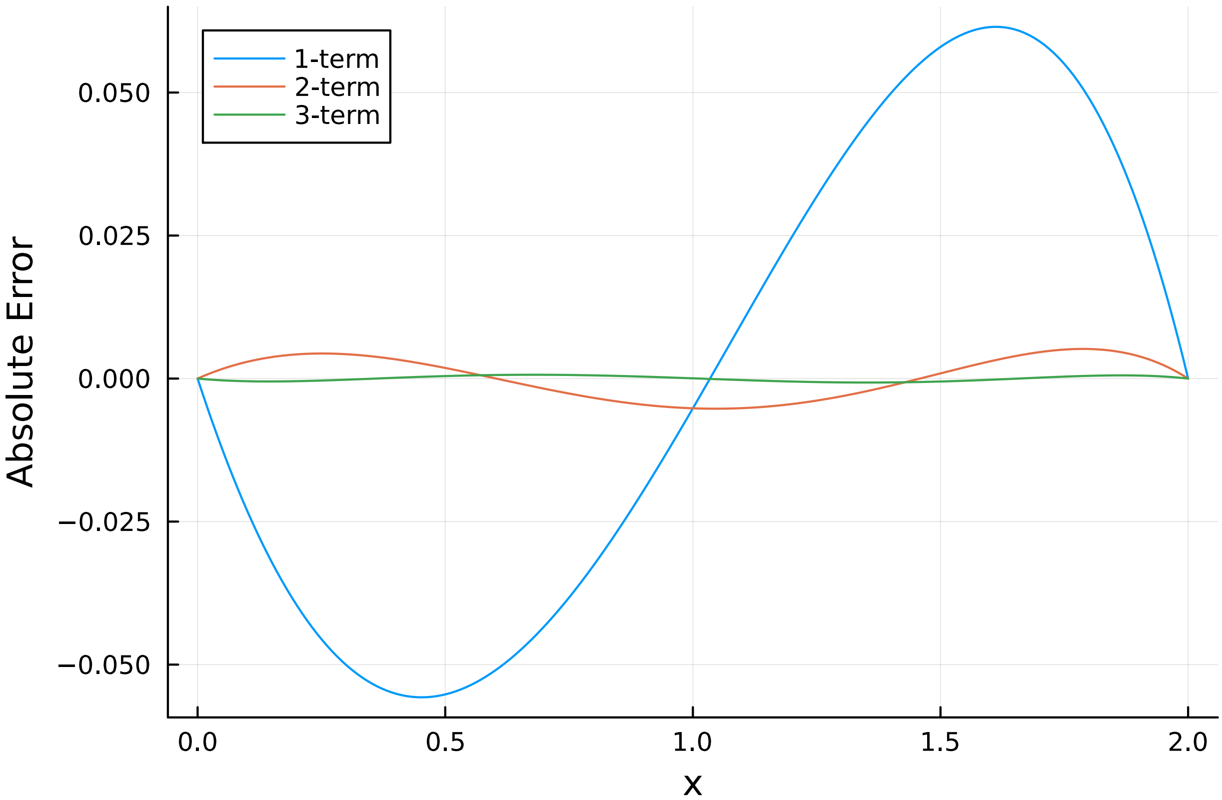
```
RitzResult:
  Domain:          [0.0, 2.0]
  Coefficients:    [-0.22826086956521813, -0.09510869565217232, -0.02853260869565301]
  Minimized value: -0.5173913043478264
  Converged:       true
```

3 Comparison

```
Term count | Minimized J[y] | Coefficients
1          | -0.47619048    | [-0.35714285714285726]
2          | -0.51677019    | [-0.20496894409937896, -0.15217391304347833]
3          | -0.51739130    | [-0.22826086956521813, -0.09510869565217232,
-0.02853260869565301]
```



4 Approximation Error


1-term
2-term
3-term
Absolute Error
0.050
0.025
0.000
−0.025
−0.050
0.0
0.5
1.0
1.5
2.0
x